

THE BLEILER–LITHERLAND CONJECTURE ON LENS SPACE SURGERIES

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ABSTRACT. It is a longstanding conjecture of Bleiler and Litherland that if surgery on a hyperbolic knot $K \subset S^3$ yields a lens space $L(r, s)$, then $r \geq 18$. Baker reduced the conjecture to excluding a hyperbolic knot $K \subset S^3$ satisfying $g(K) = 4$, $S_{14}^3(K) \cong L(14, 11)$, and $\Delta_K(t) = \Delta_{T(3,5)}(t)$. We rule out this remaining case, proving the conjecture. As a corollary, 14 is a characterizing slope for $T(3, 5)$.

1. INTRODUCTION

For a knot $K \subset S^3$, let m and l denote the meridian and preferred longitude, respectively. For coprime integers p, q , let $S_{p/q}^3(K)$ denote Dehn surgery along the slope $pm + ql$. A lens space surgery is a nontrivial surgery for which $S_{p/q}^3(K)$ is a lens space. We write a lens space as $L(r, s)$ with $r > 0$, so that $r = |H_1(L(r, s); \mathbb{Z})|$, and refer to r as the *order* of $L(r, s)$.

The study of lens space surgeries begins with Moser’s classical work on torus knots [19]. A fundamental hyperbolic example was found by Fintushel and Stern: the $(-2, 3, 7)$ -pretzel knot admits lens space surgeries of orders 18 and 19 [8]. Berge later constructed twelve families of doubly primitive knots admitting integral lens space surgeries and conjectured that these families account for all such surgeries in S^3 [3].

For hyperbolic knots, the cyclic surgery theorem of Culler, Gordon, Luecke, and Shalen implies that every lens space surgery slope is integral [7, Corollary 1].

Monopole and Heegaard Floer theories impose further strong restrictions. Kronheimer, Mrowka, Ozsváth and Szabó proved that $\mathbb{RP}^3$ cannot be obtained by surgery on a nontrivial knot and derived additional constraints on lens space surgeries [15]. Ozsváth and Szabó constrained the Alexander polynomial of a knot admitting a lens space surgery [23]. Goda–Teragaito obtained genus bounds for lens space surgery slopes [11], which Rasmussen sharpened to $|p| \leq 4g(K) + 3$ for a lens space surgery of slope p [24]. Moreover,

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Ni's fiberedness theorem shows that any such knot is fibered [21]. Greene solved the lens space realization problem: every lens space obtained by integral surgery on a knot in S^3 is also obtained by surgery on a Berge knot with the same knot Floer homology [12].

Recent progress on Berge's conjecture includes the work of Li, Moriah, and Pinsky, who proved it for tunnel-number-one knots [17].

In 1989, Bleiler and Litherland formulated the following conjecture [4].

Conjecture 1.1 (Bleiler–Litherland). *Every lens space obtained by surgery on a hyperbolic knot in S^3 has order at least 18.*

The Fintushel–Stern example shows that this bound is sharp. Baker showed that order 14 is the only remaining possibility below 18 [1, Theorem 1.6]. In the final paragraph of Section 1.3, he identified the sole unresolved case as whether ± 14 -surgery on a genus-four hyperbolic knot K with $\Delta_K(t) = \Delta_{T(3,5)}(t)$ can yield $L(14, 11)$ [1, Section 1.3, final paragraph]. After replacing K by its mirror if necessary, it therefore suffices to rule out a hyperbolic knot, still denoted K , satisfying

$$(1) \quad S_{14}^3(K) \cong L(14, 11), \quad g(K) = 4, \quad \Delta_K(t) = \Delta_{T(3,5)}(t).$$

Here and below, Alexander polynomials are symmetrically normalized; thus

$$(2) \quad \Delta_{T(3,5)}(t) = t^4 - t^3 + t - 1 + t^{-1} - t^{-3} + t^{-4} = t^{-4}\Phi_{15}(t).$$

We rule out (1), thereby resolving Baker's remaining case and proving the following theorem.

Theorem 1.2. *The Bleiler–Litherland conjecture holds.*

The lens space surgery condition first implies that K is fibered. Since K is hyperbolic, its monodromy is pseudo-Anosov, and the proof studies the stable invariant foliation of a pseudo-Anosov representative. General constraints on its boundary behavior leave only two possible boundary-prong numbers, 13 and 14. Ni's fixed-point theorem excludes the 13-prong case. In the remaining case, a mod-2 homological argument shows that the invariant foliations become orientable after the boundary is capped off. The monodromy action on first homology would then have a real eigenvalue of absolute value greater than one, contradicting the fact that $\Delta_K(t) = t^{-4}\Phi_{15}(t)$ has all its roots on the unit circle.

Remark 1.3. The hypotheses in (1) enter at different stages. The lens space surgery condition makes K an L-space knot and hence supplies fiberedness and right-veering monodromy. Together with the genus assumption $g(K) = 4$, the resulting dynamical restrictions lead to the 13- and 14-prong alternatives above. The identity $\Delta_K = \Delta_{T(3,5)}$ is used only in the final homological contradiction in the 14-prong case.

A slope p/q is called a *characterizing slope* for a knot J if an orientation-preserving homeomorphism

$$S_{p/q}^3(K) \cong S_{p/q}^3(J)$$

forces $K = J$. Moser’s surgery formula, together with the classification of lens spaces, gives

$$S_{14}^3(T(3, 5)) \cong L(14, 9) \cong L(14, 11);$$

see [19, Propositions 3.1–3.2] and [5]. These facts, together with Theorem 1.2, yield the following characterizing-slope result.

Corollary 1.4. *If*

$$S_{14}^3(K) \cong S_{14}^3(T(3, 5)),$$

then $K = T(3, 5)$.

Section 2 collects the necessary background, Section 3 proves the orientability criterion, and Section 4 completes the proof and establishes Corollary 1.4.

2. PRELIMINARIES

This section fixes our notation and records the background results used in the proof of the main theorem.

2.1. L-space knots and monodromy.

Definition 2.1. A rational homology sphere Y is an *L-space* if

$$\text{rank } \widehat{HF}(Y) = |H_1(Y; \mathbb{Z})|.$$

A knot $K \subset S^3$ is an *L-space knot* if some positive surgery on K is an L-space.

Lens spaces are L-spaces [23]. We will repeatedly use two standard consequences of the L-space surgery condition. Ni proved that the knot is fibered [21, Corollary 1.2]; his work, together with that of Honda–Kazez–Matić, also implies that the monodromy is right-veering [20, Remark 1.2] and [13, Theorem 1.1]. We record both conclusions together.

Proposition 2.2. *If a knot $K \subset S^3$ admits a positive lens space surgery, then K is fibered and its monodromy is right-veering.*

For hyperbolic L-space knots, we also need Ni’s fixed-point theorem [22, Theorem 1.1 and the paragraph following it].

Theorem 2.3. *Let $K \subset S^3$ be a hyperbolic L-space knot. Then the monodromy is freely isotopic to a pseudo-Anosov homeomorphism with no fixed points.*

Remark 2.4. The pseudo-Anosov representative supplied by Theorem 2.3 is freely isotopic to the boundary-fixing monodromy, but it need not fix the boundary pointwise.

2.2. Pseudo-Anosov dynamics on the fiber. Let F be a compact oriented surface with one boundary component, and let $h : F \rightarrow F$ fix ∂F pointwise. Suppose that the mapping class of h is pseudo-Anosov, and choose a pseudo-Anosov representative ψ freely isotopic to h . Let $(\mathcal{F}^s, \mu^s)$ and $(\mathcal{F}^u, \mu^u)$ be its stable and unstable invariant measured foliations, and let $\lambda > 1$ be its dilatation. Only the prong data and orientability of these foliations will be needed. The stable and unstable foliations have the same interior singularities, with the same prong numbers, and the same number of boundary prongs; see Baldwin–Hu–Sivek [2, Section 2].

In the standard boundary-singularity local model, a *boundary prong* is a separatrix branch incident to ∂F . Each boundary component of a pseudo-Anosov invariant foliation carries a nonempty finite set of such prongs. Because F has only one boundary component, the stable and unstable foliations have a common boundary-prong number $n \geq 1$; see Baldwin–Hu–Sivek [2, Section 2]. This local model is shown in Figure 1(a): the small circle represents ∂F , the dashed separatrix branches terminating on it are the boundary prongs, and the solid curves are nearby leaves of $\mathcal{F}^s$.

The boundary-prong data also describe the fractional Dehn twist coefficient. Let $c(h)$ denote the fractional Dehn twist coefficient of h with respect to ∂F . If the stable invariant foliation has n boundary prongs, then

$$(3) \quad c(h) = \frac{k}{n}$$

for some $k \in \mathbb{Z}$; the fraction k/n need not be reduced. Here the denominator n is the boundary-prong number; see Honda–Kazez–Matić [13, Section 3.2]. We will use this description in Subsection 2.3, together with the Honda–Kazez–Matić criterion that a pseudo-Anosov mapping class is right-veering if and only if its fractional Dehn twist coefficient is positive [13, Proposition 3.1].

For a fibered knot, the Alexander polynomial is determined by the action of the monodromy on first homology. We use the following standard formula; see Milnor [18].

Proposition 2.5. *Let K be a fibered knot with fiber F of genus g , and let $h_* : H_1(F; \mathbb{Z}) \rightarrow H_1(F; \mathbb{Z})$ be the homological monodromy. For the symmetrically normalized Alexander polynomial,*

$$(4) \quad \det(tI - h_*) = \varepsilon t^g \Delta_K(t), \quad \varepsilon \in \{1, -1\}.$$

In particular,

$$(5) \quad \det(I - h_*) = \pm \Delta_K(1).$$

We next note a simple consequence of the fixed-point-free hypothesis.

Lemma 2.6. *If a pseudo-Anosov homeomorphism $\psi : F \rightarrow F$ has no fixed points, then its boundary-prong number satisfies $n \geq 2$.*

Proof. Every pseudo-Anosov invariant foliation on F has at least one boundary prong, so $n \geq 1$. If $n = 1$, then ψ must preserve the unique boundary prong and hence fix its endpoint on ∂F , a contradiction. $\square$

We now assume $n \geq 2$ and cap off the boundary. Let

$$q : F \longrightarrow \widehat{F} = F/\partial F, \quad v = q(\partial F)$$

be the quotient collapsing ∂F to a point. Panel (b) of Figure 1 shows the resulting local picture. The endpoints of the boundary prongs in panel (a) are identified at v , and the prongs become the separatrices issuing from v in panel (b). Thus, in the four-prong model shown, v is a four-pronged singularity of the induced foliation $\widehat{\mathcal{F}}^s$. More generally, n boundary prongs give an n -pronged point after capping; the point is regular when $n = 2$ and singular when $n > 2$. Away from ∂F , the foliation is unchanged. The same description applies to the unstable foliation.

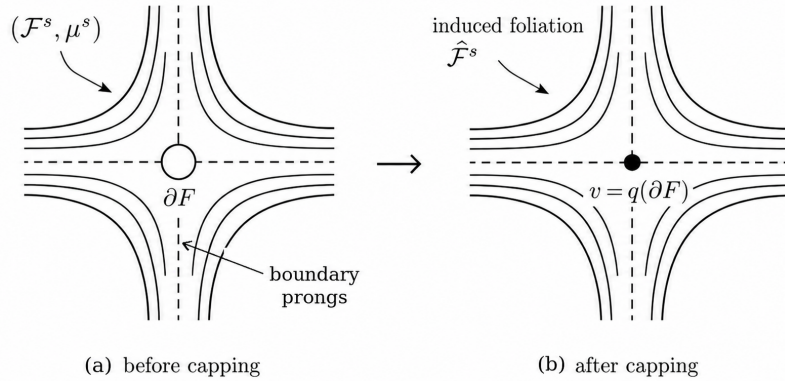


FIGURE 1. Local capping model for the stable foliation in the case $n = 4$. Panel (a) shows four boundary prongs terminating on ∂F . Collapsing ∂F to $v = q(\partial F)$ identifies their endpoints and produces the four-pronged singularity of $\widehat{\mathcal{F}}^s$ shown in panel (b).

When $n \geq 2$, this construction preserves the pseudo-Anosov dynamics in the following standard sense.

Lemma 2.7. *Assume that ψ has $n \geq 2$ boundary prongs. Then ψ induces a pseudo-Anosov homeomorphism*

$$\widehat{\psi} : \widehat{F} \longrightarrow \widehat{F}$$

with the same dilatation λ . Its invariant measured foliations are obtained by extending those of ψ across v : the n boundary prongs become an n -pronged singularity at v when $n > 2$, while v is regular when $n = 2$.

Proof. This is the standard capping construction for a pseudo-Anosov map with at least two boundary prongs; see Baldwin–Hu–Sivek [2, Section 2]. Because ψ preserves ∂F setwise, it descends to a homeomorphism $\widehat{\psi}$ fixing v . The invariant measured foliations extend across v through the standard

n -prong local model. If μ^s and μ^u denote their transverse measures, then ψ scales them by λ^{-1} and λ , respectively. The extended measures agree with μ^s and μ^u away from v , and the point v contributes no transverse measure. Hence the same scaling relations hold after capping, and $\hat{\psi}$ is pseudo-Anosov with dilatation λ . $\square$

The Euler–Poincaré formula on the capped surface constrains the boundary-prong number. We record the form needed later.

Lemma 2.8. *Suppose that the stable invariant foliation $\mathcal{F}^s$ has $n \geq 2$ boundary prongs and that its interior singularities have $p_1, \dots, p_N$ prongs. Then*

$$(6) \quad (n-2) + \sum_{i=1}^N (p_i - 2) = 4g(F) - 4.$$

Proof. By Lemma 2.7, the stable foliation extends to the closed surface $\hat{F}$ of genus $g(F)$. The capped point is an n -pronged singularity when $n \geq 3$, and is a regular point when $n = 2$; the remaining singularities have prong numbers $p_1, \dots, p_N$. The standard Euler–Poincaré prong formula on a closed surface therefore gives

$$(n-2) + \sum_{i=1}^N (p_i - 2) = 4g(\hat{F}) - 4.$$

Since $g(\hat{F}) = g(F)$, this is exactly (6). $\square$

Recall that a singular measured foliation is *orientable* if the tangent lines to its leaves can be oriented continuously on the complement of the singular set. Equivalently, its tangent line bundle on the nonsingular locus is orientable. In Section 3 we use the following standard formulation in terms of the first Stiefel–Whitney class, together with the local criterion at a pronged singularity.

Lemma 2.9. *Let $\mathcal{F}$ be a singular measured foliation on a surface S , let Σ be its singular set, put $U = S \setminus \Sigma$, and let $L = T\mathcal{F}|_U$ be its tangent line bundle. Then*

$$\mathcal{F} \text{ is orientable} \iff w_1(L) = 0 \text{ in } H^1(U; \mathbb{F}_2).$$

Moreover, if $z \in \Sigma$ is a p -pronged singularity and γ_z is a small loop around z , then

$$\langle w_1(L), [\gamma_z] \rangle \equiv p \pmod{2}.$$

Thus the local orientation obstruction at z vanishes if and only if p is even.

When an invariant measured foliation is orientable, the pseudo-Anosov dilatation occurs, up to sign, as an eigenvalue of the action on first homology; we record this standard fact in the following proposition.

Proposition 2.10. *Let $\psi : F \rightarrow F$ be pseudo-Anosov with $n \geq 2$ boundary prongs and dilatation $\lambda > 1$. Suppose that, after capping ∂F , one of the invariant measured foliations of the induced map $\hat{\psi}$ is orientable. Then the maps*

$$\hat{\psi}_* : H_1(\hat{F}; \mathbb{R}) \longrightarrow H_1(\hat{F}; \mathbb{R}) \quad \text{and} \quad \psi_* : H_1(F; \mathbb{R}) \longrightarrow H_1(F; \mathbb{R})$$

have a common real eigenvalue ρ satisfying $|\rho| = \lambda > 1$.

Proof. By Lemma 2.7, $\hat{\psi}$ is pseudo-Anosov on the closed surface $\hat{F}$ and has the same dilatation λ . Since $\hat{\psi}$ leaves invariant an orientable measured foliation, Thurston’s homological theorem, in the form stated by Lanneau–Thiffeault [16, Theorem 2.2], gives a real eigenvalue ρ of

$$\hat{\psi}_* : H_1(\hat{F}; \mathbb{R}) \longrightarrow H_1(\hat{F}; \mathbb{R})$$

with $|\rho| = \lambda > 1$.

Because F has one boundary component, the quotient map induces an isomorphism $q_* : H_1(F; \mathbb{R}) \rightarrow H_1(\hat{F}; \mathbb{R})$. Naturality gives the commutative diagram

$$\begin{array}{ccc} H_1(F; \mathbb{R}) & \xrightarrow[\cong]{\psi_*} & H_1(F; \mathbb{R}) \\ q_* \downarrow \cong & & \cong \downarrow q_* \\ H_1(\hat{F}; \mathbb{R}) & \xrightarrow[\cong]{\hat{\psi}_*} & H_1(\hat{F}; \mathbb{R}). \end{array}$$

Since the vertical maps are isomorphisms, the two homological actions are conjugate under q_* . Thus ψ_* has the same eigenvalue ρ , as claimed. $\square$

2.3. Stable laminations and Dehn filling. We now turn from the invariant foliations on the fiber to the associated lamination in the knot exterior. Let $K \subset S^3$ be a hyperbolic fibered knot with exterior X_K , and let $\lambda^s \subset \text{int } X_K$ denote the stable lamination of its pseudo-Anosov monodromy. Here λ^s is a two-dimensional lamination in the three-manifold X_K , rather than a one-dimensional foliation of the fiber. In particular,

$$\lambda^s \cap \partial X_K = \emptyset.$$

The lamination λ^s is transverse to the fibration, very full, and essential. Only its degeneracy slope and its behavior under Dehn filling will be needed; see Ni [20, Section 2.1] and Calegari [6, Example 6.37 and the discussion preceding Definition 6.42].

The boundary behavior of λ^s determines a primitive degeneracy slope. In the meridian–longitude coordinates fixed in the Introduction, write

$$\gamma(\lambda^s) = \frac{A}{B}, \quad \gcd(A, B) = 1.$$

The reciprocal relation with the fractional Dehn twist coefficient and Gabai’s degeneracy-slope bound give the following restriction.

Proposition 2.11. *Let $K \subset S^3$ be a hyperbolic fibered knot of genus g whose monodromy h is right-veering. Then, for the stable lamination λ^s ,*

$$\gamma(\lambda^s) = \frac{d}{1}, \quad 2 \leq d \leq 4g - 2, \quad c(h) = \frac{1}{d}.$$

Proof. By Gabai's degeneracy-slope theorem, $\gamma(\lambda^s)$ is either $1/0$ or $d/1$, with $2 \leq |d| \leq 4g - 2$ in the latter case [9, Theorem 8.8]. The reciprocal relation gives $c(h) = 0$ in the first case and $c(h) = 1/d$ in the second; see Ni [20, Proposition 2.3 and Theorem 2.4]. Since K is hyperbolic and fibered, the mapping class of h is pseudo-Anosov; hence the right-veering criterion recalled above gives $c(h) > 0$. Thus the first case is impossible and $d > 0$, proving the proposition. $\square$

Since $\lambda^s \subset \text{int } X_K$, it also determines a lamination in the filled manifold $S^3_{p/q}(K)$. Its essentiality there, however, is not automatic. For a primitive filling slope p/q , define its distance from the degeneracy slope $\gamma(\lambda^s) = A/B$ by

$$\Delta\left(\gamma(\lambda^s), \frac{p}{q}\right) = |Aq - Bp|.$$

We will use the following persistence result.

Proposition 2.12. *Let $K \subset S^3$ be a hyperbolic fibered knot and let λ^s be its stable lamination. If*

$$\Delta\left(\gamma(\lambda^s), \frac{p}{q}\right) \geq 2,$$

then λ^s remains an essential lamination in $S^3_{p/q}(K)$. Consequently, the universal cover of $S^3_{p/q}(K)$ is homeomorphic to $\mathbb{R}^3$, its fundamental group is infinite, and $S^3_{p/q}(K)$ is not a lens space.

Proof. The first assertion is the persistence theorem of Gabai–Oertel [10, Theorem 5.3]; see also Calegari [6, Theorem 6.48] and Ni [20, Theorem 2.5]. By Gabai–Oertel [10, Theorem 6.1], a closed 3-manifold containing an essential lamination has universal cover $\mathbb{R}^3$; in particular, its fundamental group is infinite. The last assertion follows because a lens space has finite fundamental group. $\square$

2.4. Torus-knot lens space surgeries. Moser's classification gives the following criterion in our convention [19, Propositions 3.1–3.2].

Theorem 2.13 (Moser). *Let $r > s > 1$ be coprime integers. Surgery on the torus knot $T(r, s)$ with integer coefficient N is a lens space if and only if*

$$|N - rs| = 1.$$

For the mirror $\overline{T(r, s)}$, the corresponding condition is $|N + rs| = 1$.

Corollary 2.14. *Among nontrivial torus knots and their mirrors, the only knot admitting 14-surgery to a lens space is $T(3, 5)$.*

Proof. For the mirror, $r > s > 1$ implies $rs \geq 6$, and hence

$$|14 + rs| = 14 + rs \geq 20 > 1,$$

so the mirror case is impossible. For $T(r, s)$, Moser's condition gives

$$|14 - rs| = 1,$$

so $rs = 13$ or 15 . The first possibility is impossible because $r, s > 1$. The second, together with $r > s > 1$ and $\gcd(r, s) = 1$, gives $(r, s) = (5, 3)$. Since $T(5, 3) = T(3, 5)$, the result follows. $\square$

3. A MOD-2 HOMOLOGICAL CRITERION FOR ORIENTABILITY

The 14-prong case of the main argument depends on the orientability of the capped invariant foliations. Even prongs eliminate the local orientation obstructions, but a global obstruction measured by the first Stiefel–Whitney class may remain. The next proposition shows that the monodromy action on mod-2 homology eliminates this obstruction.

Proposition 3.1 (Mod-2 orientability criterion). *Let F be a compact oriented surface with one boundary component, and let $h : F \rightarrow F$ be a boundary-fixing homeomorphism whose mapping class is pseudo-Anosov. Let $\psi : F \rightarrow F$ be a pseudo-Anosov homeomorphism freely isotopic to h , with invariant foliations $\mathcal{F}^s$ and $\mathcal{F}^u$. Suppose that every interior singularity has an even number of prongs and that the boundary-prong number n is even. Denote the induced maps by*

$$h_{*,2} : H_1(F; \mathbb{F}_2) \longrightarrow H_1(F; \mathbb{F}_2)$$

and

$$h_{*,\mathbb{Z}} : H_1(F; \mathbb{Z}) \longrightarrow H_1(F; \mathbb{Z}).$$

- (1) *If $I - h_{*,2}$ is an isomorphism, then the invariant foliations extend to orientable foliations on the capped surface $\widehat{F}$.*
- (2) *In particular, the same conclusion holds if $\det(I - h_{*,\mathbb{Z}})$ is odd.*

Proof. We begin with (1), taking all homology and cohomology groups with coefficients in $\mathbb{F}_2$. Since n is positive and even, $n \geq 2$. By Lemma 2.7, ψ induces a pseudo-Anosov homeomorphism

$$\widehat{\psi} : \widehat{F} \longrightarrow \widehat{F}.$$

Let $\widehat{\mathcal{F}}^s$ and $\widehat{\mathcal{F}}^u$ denote the capped invariant foliations. If $\widehat{\Sigma}$ is their common singular set, put

$$U = \widehat{F} \setminus \widehat{\Sigma}.$$

Let $i : U \hookrightarrow \widehat{F}$ be the inclusion, and let $q : F \rightarrow \widehat{F}$ collapse ∂F to a point. The following diagram

$$\begin{array}{ccccccc} U & \xrightarrow{i} & \widehat{F} & \xleftarrow{q} & F \\ \widehat{\psi}|_U \downarrow & & \downarrow \widehat{\psi} & & \downarrow \psi \\ U & \xrightarrow{i} & \widehat{F} & \xleftarrow{q} & F \end{array}$$

commutes. Passing to homology gives

$$\begin{array}{ccccc} H_1(U) & \xrightarrow{i_*} & H_1(\widehat{F}) & \xleftarrow[\cong]{q_*} & H_1(F) \\ (\widehat{\psi}|_U)_* \downarrow \cong & & \cong \downarrow \widehat{\psi}_* & & \cong \downarrow \psi_* \\ H_1(U) & \xrightarrow{i_*} & H_1(\widehat{F}) & \xleftarrow[\cong]{q_*} & H_1(F). \end{array}$$

Here q_* is an isomorphism because F has one boundary component. Dualizing gives the commutative diagram

$$\begin{array}{ccccc} H^1(U) & \xleftarrow{i^*} & H^1(\widehat{F}) & \xrightarrow[\cong]{q^*} & H^1(F) \\ (\widehat{\psi}|_U)^* \uparrow \cong & & \cong \uparrow \widehat{\psi}^* & & \cong \uparrow \psi^* \\ H^1(U) & \xleftarrow{i^*} & H^1(\widehat{F}) & \xrightarrow[\cong]{q^*} & H^1(F). \end{array}$$

Choose disjoint small disk neighborhoods of the points in $\widehat{\Sigma}$, and let γ_z be a loop around the puncture corresponding to z . The Mayer–Vietoris sequence for adjoining these disks to U contains the exact sequence

$$\bigoplus_{z \in \widehat{\Sigma}} \mathbb{F}_2[\gamma_z] \longrightarrow H_1(U) \xrightarrow{i_*} H_1(\widehat{F}) \longrightarrow 0.$$

Thus i_* is surjective. Dualizing over $\mathbb{F}_2$ gives

$$(7) \quad 0 \longrightarrow H^1(\widehat{F}) \xrightarrow{i^*} H^1(U) \longrightarrow \text{Hom} \left(\bigoplus_{z \in \widehat{\Sigma}} \mathbb{F}_2[\gamma_z], \mathbb{F}_2 \right).$$

In particular, i^* is injective.

Let $L = T\widehat{\mathcal{F}}^s|_U \subset TU$ be the tangent line bundle of the stable foliation on the nonsingular locus. Its first Stiefel–Whitney class is

$$w_1(L) \in H^1(U).$$

Every singularity of $\widehat{\mathcal{F}}^s$ is even-pronged. Indeed, the interior singularities are even-pronged by hypothesis, while the capped boundary point is n -pronged if $n > 2$ and regular if $n = 2$. Hence Lemma 2.9 gives

$$\langle w_1(L), [\gamma_z] \rangle = 0 \quad (z \in \widehat{\Sigma}).$$

Thus $w_1(L)$ maps to zero under the right-hand map in (7). By exactness, there is a class $w \in H^1(\widehat{F})$ such that

$$i^*w = w_1(L).$$

The injectivity of i^* makes this class unique.

Since $\widehat{\psi}$ preserves $\widehat{\mathcal{F}}^s$, the invariance of the foliation identifies L with $(\widehat{\psi}|_U)^*L$. Naturality of the first Stiefel–Whitney class therefore gives

$$(\widehat{\psi}|_U)^*w_1(L) = w_1((\widehat{\psi}|_U)^*L) = w_1(L).$$

Using $i^*w = w_1(L)$ and the commutativity of the cohomology diagram, we obtain

$$\begin{aligned} i^*(\widehat{\psi}^*w - w) &= (\widehat{\psi}|_U)^*i^*w - i^*w \\ &= (\widehat{\psi}|_U)^*w_1(L) - w_1(L) = 0. \end{aligned}$$

Since i^* is injective,

$$\widehat{\psi}^*w = w, \quad \text{or equivalently} \quad (I - \widehat{\psi}^*)w = 0.$$

Moreover, $\psi_* = h_{*,2}$ because h and ψ are freely isotopic. The right-hand square of the homology diagram therefore shows that

$$I - \widehat{\psi}_* = q_*(I - h_{*,2})q_*^{-1}.$$

By hypothesis, $I - \widehat{\psi}_*$ is invertible; hence its dual $I - \widehat{\psi}^*$ is also invertible. The invariant class w must therefore vanish, and so $w_1(L) = 0$. Hence $\widehat{\mathcal{F}}^s$ is orientable, and the same argument applies to $\widehat{\mathcal{F}}^u$.

For (2), choose an integral basis of $H_1(F; \mathbb{Z})$. Reduction modulo 2 gives

$$\det(I - h_{*,2}) \equiv \det(I - h_{*,\mathbb{Z}}) \pmod{2}.$$

If $\det(I - h_{*,\mathbb{Z}})$ is odd, then $\det(I - h_{*,2}) = 1$ in $\mathbb{F}_2$. Therefore $I - h_{*,2}$ is an isomorphism, and part (1) applies. $\square$

4. EXCLUDING THE 14-SURGERY CASE

We now return to Baker's remaining case and use the preceding results to rule out the two extremal boundary-prong configurations that arise.

Proposition 4.1. *There is no hyperbolic knot $K \subset S^3$ of genus four with $\Delta_K(t) = \Delta_{T(3,5)}(t)$ such that $S_{14}^3(K)$ is a lens space.*

Proof. Suppose, to the contrary, that such a knot exists. Since $S_{14}^3(K)$ is a lens space, K is an L-space knot. Proposition 2.2 shows that K is fibered and that its monodromy h is right-veering. Let ψ be the fixed-point-free pseudo-Anosov representative supplied by Theorem 2.3, and let n be its boundary-prong number. By Lemma 2.6, $n \geq 2$. Since $g(F) = 4$, Lemma 2.8 gives

$$(n - 2) + \sum_{i=1}^N (p_i - 2) = 12.$$

Every interior singularity has at least three prongs, so the sum is nonnegative. It follows that $n \leq 14$, and hence

$$2 \leq n \leq 14.$$

We next restrict the degeneracy slope. Since $g(K) = 4$, Proposition 2.11 gives

$$(8) \quad \gamma(\lambda^s) = \frac{d}{1}, \quad 2 \leq d \leq 4g(K) - 2 = 14.$$

If $d \leq 12$, then $\Delta(d/1, 14/1) = 14 - d \geq 2$, and Proposition 2.12 would imply that $S_{14}^3(K)$ is not a lens space, a contradiction. Hence $d \in \{13, 14\}$. We now compare n with d . By (3) and Proposition 2.11,

$$\frac{k}{n} = c(h) = \frac{1}{d}$$

for some positive integer k . Hence $n = kd$, and in particular $d \mid n$. Combining this divisibility with $d \in \{13, 14\}$ and $n \leq 14$ leaves only

$$(d, n) = (13, 13) \quad \text{or} \quad (d, n) = (14, 14).$$

Consider first the case $d = n = 13$. Equation (6) becomes

$$(13 - 2) + \sum_{i=1}^N (p_i - 2) = 4 \cdot 4 - 4 = 12,$$

and hence

$$\sum_{i=1}^N (p_i - 2) = 1.$$

Thus there is exactly one interior singularity, and it is three-pronged. Although ψ may permute its prongs, it must fix the singularity itself, contradicting Theorem 2.3.

It remains to consider $d = n = 14$. In this case,

$$(14 - 2) + \sum_{i=1}^N (p_i - 2) = 12,$$

and therefore

$$\sum_{i=1}^N (p_i - 2) = 0.$$

Thus there are no interior singularities, and the boundary-prong number $n = 14$ is even. The even-prong hypotheses of Proposition 3.1 are thus satisfied. Moreover, Proposition 2.5 and (2) give

$$\Delta_K(1) = \Delta_{T(3,5)}(1) = 1, \quad \det(I - h_*) = \pm 1,$$

which is odd. Proposition 3.1(2) now implies that the invariant foliations extend orientably over the capped surface. By Proposition 2.10, the map ψ_* has a real eigenvalue ρ with $|\rho| = \lambda > 1$. Since h and ψ are freely isotopic, they induce the same map on homology, so ρ is also an eigenvalue of h_* . On the other hand, (4) and (2) give

$$\det(tI - h_*) = \varepsilon \Phi_{15}(t).$$

Every eigenvalue of h_* is therefore a root of Φ_{15} and has absolute value one, a contradiction. $\square$

Proof of Theorem 1.2. Suppose that surgery on a hyperbolic knot yields a lens space $L(r, s)$ with $r < 18$. Baker’s Theorem 1.6 gives $r = 14$, and the final paragraph of his Section 1.3 identifies the sole unresolved case. After replacing the knot by its mirror if necessary, one is therefore in the situation of (1); see [1, Theorem 1.6 and Section 1.3, final paragraph]. Proposition 4.1 rules out that case. Hence $r \geq 18$. $\square$

Proof of Corollary 1.4. By the identification in the Introduction, the common surgery manifold is homeomorphic to $L(14, 11)$. If K is the unknot, then $S^3_{14}(K) = L(14, 1)$, which is not homeomorphic to $L(14, 11)$ by the classification of lens spaces [5].

Thus K is nontrivial. Since the 14-surgery is an L-space, K is an L-space knot, and Krcatovich [14, Theorem 1.2] implies that K is prime. Accordingly, K is a torus knot, a satellite knot, or a hyperbolic knot. If K is a torus knot, Corollary 2.14 gives $K = T(3, 5)$, as required. If K is a satellite knot, Bleiler–Litherland [4, Corollary] show that a lens space obtained by surgery on a satellite knot satisfies $|\pi_1| \geq 23$. This is impossible here, since the surgery manifold has $|\pi_1| = 14$. Finally, if K is hyperbolic, Theorem 1.2 excludes this case. $\square$

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